# Data Flow Diagrams & User Stories

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| **Date** | 31 January 2026 |
| **Team ID** | Not provided — confirm with MySkillWallet/college |
| **Project Name** | EV Market Intelligence Dashboard |
| **Maximum Marks** | 4 Marks |

## Data Flow Diagram (Level 0)

CSV dataset → Tableau Public data source → field validation / aggregation / worksheet creation → nine-view dashboard → Tableau Public link and project documentation. The project is a static analytical dashboard, not a live operational data pipeline.

## DFD entities and processes

| Type | Project-specific element | Description |
|---|---|---|
| External entity | EV market CSV dataset | Source contains vehicle category, manufacturer, country, battery, range, sales, rating, model-year and sustainability fields. |
| Process 1 | Load data | Connect the CSV file to Tableau Public. |
| Process 2 | Prepare and analyze | Check field types; aggregate sales and ratings; create category, manufacturer, battery, range, map, model-year and sustainability views. |
| Data store | Tableau workbook/data source | The workbook stores worksheet definitions and the configured data connection; the local project files are retained for submission. |
| External entity | Dashboard users | Emily Johnson, David Chen and Dr. Sarah Williams explore the views to answer their scenario questions. |
| Output | Published dashboard and documentation | Share the Tableau Public URL and provide workbook, completed templates, and presentation video. |

## User Stories

| User Type | Functional Requirement (Epic) | User Story Number | User Story / Task | Acceptance criteria | Priority | Release |
|---|---|---|---|---|---|---|
| Market Strategy Director | Sales & Category Analysis | USN-1 | As a market strategy director, I can compare annual sales across EV categories. | Categories are ranked by total annual sales and values are visible in tooltips/labels. | High | Sprint-1 |
| BI Manager | Manufacturer Performance | USN-2 | As a BI manager, I can compare manufacturers by annual sales and identify leading competitors. | Manufacturers can be sorted by sales and the top 10 can be reviewed clearly. | High | Sprint-1 |
| Sustainability Research Analyst | Battery Technology Mix | USN-3 | As a research analyst, I can view the distribution of battery types in the dataset. | Each battery type has a distinct legend/color and the share/count is understandable. | High | Sprint-1 |
| Market Strategy Director | Driving Range Comparison | USN-4 | As a market strategy director, I can compare driving range across vehicle categories. | The view compares range by category and uses consistent units. | High | Sprint-1 |
| BI Manager | Manufacturing Geography | USN-5 | As a BI manager, I can explore EV manufacturing countries and their associated sales. | The map identifies available manufacturing countries and provides sales context on hover. | Medium | Sprint-2 |

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|---|---|---|---|---|---|---|
| Sustainability Research Analyst | Customer Satisfaction | USN-6 | As a researcher, I can compare average customer ratings across vehicle categories. | Average rating is shown by category and can be compared without misleading scales. | High | Sprint-1 |
| Market Strategy Director | Model-Year Sales Trend | USN-7 | As a market strategy director, I can inspect annual sales by model year. | Sales are plotted by model year to expose changes and outliers in the dataset. | Medium | Sprint-2 |
| Sustainability Research Analyst | Battery Capacity vs Range | USN-8 | As a researcher, I can examine the relationship between battery capacity and driving range. | A scatter plot shows capacity against range and distinguishes battery technologies. | High | Sprint-1 |
| Sustainability Research Analyst | Sustainability Impact | USN-9 | As a researcher, I can compare estimated CO■ savings across vehicle categories. | CO■ saved per year is compared by category with the measure's unit shown. | Medium | Sprint-2 |
| Dashboard User | Interactive Filtering | USN-10 | As a dashboard user, I can filter or highlight relevant categories and manufacturers. | Filters/highlighting update relevant views and do not obscure the chart meaning. | Medium | Sprint-2 |
| Administrator / Project Owner | Publication & Sharing | USN-11 | As the project owner, I can publish the dashboard and share a stable public link. | The published dashboard opens and all nine visualizations are readable at the chosen view size. | High | Sprint-2 |
| Dashboard User | Data Interpretation | USN-12 | As a dashboard user, I can inspect a mark's details through its tooltip. | Tooltips show relevant dimensions and measures with clear labels and units. | Medium | Sprint-2 |